

LRSC v1.1.1

Complete $K_{0.001} = 11$ Reproducibility Supplement

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Purpose. This supplement provides executable source and numerical evidence for the fixed single-step benchmark reported as $K_{0.001}(0.50) = 11$. It preserves the historical v1.1 and v1.2 records; it does not replace or revise the OSF record DOI 10.17605/OSF.IO/NM5BW.

Result at a glance

Statement	Reproduced value
Operator closure, all 50 channels	$2.3808015585084204 \times 10^{-16}$ relative L2
Best exact-cardinality $K=10$ subset	$\{0,1,42,43,44,45,46,47,48,49\}$; error 0.0010398206190990441
Passing $K=11$ witness	$\{0,1,3,42,43,44,45,46,47,48,49\}$; error 0.00096236912766441347
Numerical conclusion	$K_{0.001} = 11$ for the stated benchmark, under IEEE double-precision exhaustive search

1. Scope and definitions

The benchmark is one application of the LRSC synchronous update map to a periodic ring, with $M = 99$, input mode index $k_{\text{idx}} = 49$, threshold $\theta = 0.08$, and amplitude $A = 0.50$. The state is $X_i = 1 + A \cos(2\pi k_{\text{idx}} i/M)$, for $i = 0, \dots, M-1$. For periodic indices:

$$C_i = \frac{1}{2}[(X_i - X_{i-1})^2 + (X_i - X_{i+1})^2], \quad m_i = 1[C_i > \theta], \quad j_i = X_i - \min(X_{i-1}, X_i, X_{i+1}), \quad Y_i = m_i j_i.$$

The forward transform is normalized: $\hat{x}_q = (1/M) \sum_i x_i \exp(-2\pi i q/M)$. Set $\kappa_q = 2\pi q/M$ and $\Delta\hat{x}_q = (\cos \kappa_q - 1)\hat{y}_q$. NumPy arrays use $q = 0, \dots, 98$. The channel set is $\{0, \dots, 49\}$: channel 0 is $V_0(q) = (\cos \kappa_q - 1)\hat{j}_0 m_q$; for $r=1, \dots, 49$, $V_r(q) = (\cos \kappa_q - 1)[\hat{j}_r m_{q-r} + \hat{j}_{99-r} m_{q-(99-r)}]$, with all mask indices modulo M . DC is included in the cardinality.

Under the normalized DFT convention, the product convolution is $\hat{y}_q = \sum_r \hat{m}_{q-r} \hat{j}_r$ (no extra $1/M$ factor). The exact full-channel identity is $\Delta\hat{x} = \sum_{r=0}^{49} V_r$; closure is checked numerically before subset search.

2. Objective and exhaustive procedure

For each exact cardinality K , the program minimizes the relative complex L2 residual $\epsilon(S) = \|\Delta\hat{x} - \sum_{r \in S} V_r\|_2 / \|\Delta\hat{x}\|_2$ over all K -element subsets S of the 50 channels. It forms the equivalent real Gram objective $c + \sum_{r \in S} f_r + \sum_{\{r,s \in S\}} Q_{rs}$ by concatenating real and imaginary components, then a C++17 recursive enumerator visits each increasing index tuple exactly once. The code retains the best and runner-up scores. It verifies the best and runner-up by recomputing the residual directly from the

original complex vectors.

No LP solver or branch-and-bound relaxation is used in this supplement: all subsets for each $K=1,\dots,10$ are explicitly enumerated. $K=11$ need not be exhaustively optimized to determine the minimum cardinality, because every smaller exact cardinality fails and one $K=11$ witness passes.

3. Exhaustive results

K	Subsets visited	Best subset	Direct relative residual	≤ 0.001 ?
1	50	{49}	0.024191796924432525	No
2	1,225	{48,49}	0.021715391412619335	No
3	19,600	{0,1,49}	0.008528939402413411	No
4	230,300	{0,1,48,49}	0.0025003114814419855	No
5	2,118,760	{0,1,3,48,49}	0.002413193978566799	No
6	15,890,700	{0,1,46,47,48,49}	0.0018745070795454781	No
7	99,884,400	{0,1,45,46,47,48,49}	0.001735356145187642	No
8	536,878,650	{0,1,44,45,46,47,48,49}	0.001302595868239026	No
9	2,505,433,700	{0,1,43,44,45,46,47,48,49}	0.0011898947351241024	No
10	10,272,278,170	{0,1,42,43,44,45,46,47,48,49}	0.0010398206190990441	No

At $K=11$, the recorded witness {0,1,3,42,43,44,45,46,47,48,49} has direct relative residual 0.00096236912766441347. Thus this benchmark has no passing subset of size at most 10 and has a passing subset of size 11. The empty subset has residual 1.

4. Audit checks and reproducibility

The audit reports $N_{\text{act}} = 81$ and full-channel relative closure $2.3808015585084204 \times 10^{-16}$. It also recomputes two previously disputed $K=7$ candidates directly: {0,1,2,46,47,48,49} gives 0.0029954427926693612; {0,1,3,46,47,48,49} gives 0.0018748270325148978. Neither is the exhaustive $K=7$ minimum.

Run from the extracted source bundle:

```
python3 audit_k1_to_k10.py
```

The driver compiles `exhaustive_search.cpp` using `g++ -O3 -std=c++17 -shared -fPIC`, constructs the operator and Gram arrays in NumPy, and checks the enumerated count against `binom(50,K)` for each K . It also checks the Gram score of each reported best candidate against a direct complex-vector residual. The source output file records visited counts, best and runner-up candidates, disputed candidates, compiler-derived search results, and the $K=11$ witness.

Recorded environment: Python 3.12.14, NumPy 2.3.5, g++ 13.3.0 (Ubuntu 13.3.0-6ubuntu2~24.04), IEEE 754 binary64 arithmetic. Exact compiler and library versions can affect final low-order bits; the script prints full precision.

5. Evidence limits and version status

This is a reproducible numerical certificate, not an exact or interval proof. The combinatorial enumeration is exhaustive, but its objective is computed in floating-point arithmetic. Direct residual checks protect against a faulty candidate reconstruction; they do not provide a rigorous rounding-error enclosure for every searched subset. No claim of rational exactness, interval arithmetic, solver-independent formal

certification, or universal parameter validity is made. The conclusion applies only to the operator definitions and benchmark values in this supplement.

Versioning: v1.1 is retained as a historical release. v1.1.1 is a separately identified reproducibility supplement. The existing v1.2 material and OSF DOI 10.17605/OSF.IO/NM5BW remain frozen and are not replaced by this supplement.

Contents of the companion source bundle

audit_k1_to_k10.py (array construction, objective formation, independent direct checks); exhaustive_search.cpp (memory-safe recursive exhaustive enumeration); VERIFICATION_OUTPUT.txt (full run output); README.md (reproduction instructions and result scope); SHA256SUMS.txt (file integrity hashes). The PDF and its source are included in the repository release package.